

NORMTOKEN: SPEAKER- AND DURATION-NORMALIZED SEMANTIC SPEECH TOKENS VIA ITERATIVE S2U–T2U REFINEMENT

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ABSTRACT

Semantic speech tokens should preserve linguistic content while suppressing utterance-specific acoustic and duration variation. However, existing speech-to-unit (S2U) tokenizers often retain speaker-dependent acoustic variation and duration information. Therefore, we propose NormToken, an iterative semantic token purification framework that alternates S2U and text-to-unit (T2U) training. Each T2U model predicts text-conditioned pseudo-targets for a newly initialized S2U tokenizer, whose outputs define the target space for the next iteration. This cycle drives the two models toward a shared, text-predictable token space. Experiments on Mandarin and English show improved S2U–T2U agreement and parallel-utterance token consistency. De-tokenizers trained on the initial and refined token spaces further prove that the refined token maintains competitive WER and CER while improving speaker similarity in both voice cloning and text-to-speech synthesis. In voice cloning, speech generated with the refined tokens has a speaking rate closer to that of the acoustic reference, suggesting that the tokens retain less duration information.

Index Terms— semantic speech tokens, speaker normalization, speech tokenization, text-to-unit modeling

1. INTRODUCTION

Semantic speech tokens provide a compact interface for speech recognition, translation, and generation [1, 2]. Ideally, they should represent what is said rather than how or by whom it is spoken. However, existing speech-to-unit (S2U) tokenizers often retain non-linguistic properties of the input speech. Consequently, utterances with the same linguistic content may produce different token sequences when spoken by different speakers or at different rates. Such variation is undesirable for semantic tokens, which should primarily preserve linguistic content. ContentVec shares the motivation of suppressing speaker-dependent variation, but operates on continuous self-supervised representations [3]. In contrast, we focus on low-rate discrete tokens that can be modeled directly for speech generation. Our key observation is that linguistic content is predictable from text, whereas speaker characteristics and speaking rate are not fully specified by it. We therefore use agreement with a text-to-unit (T2U) model as a normalization signal to purify the semantic tokens generated by S2U sequences while suppressing speaker- and duration-dependent variation.

Based on this observation, we propose NormToken, an iterative semantic token purification framework. The procedure trains a T2U model to predict the deduplicated output of an initial S2U tokenizer. It then decodes text-conditioned pseudo-targets and uses them to train a newly initialized S2U model with connectionist temporal classification (CTC) [4]. The updated S2U tokenizer supplies new targets for the following T2U model, forming the ordered cycle $S_0 \rightarrow T_0 \rightarrow S_1 \rightarrow T_1 \rightarrow \dots$. Unlike using T2U merely as a data generator, NormToken transfers text-predictable structure back into the tokenizer at every iteration.

Our contributions are threefold: (i) we characterize cross-speaker sequence inconsistency and source-duration dependence in discrete semantic tokens; (ii) we propose NormToken, an iterative S2U–T2U procedure that transfers text-predictable structure back to the speech tokenizer; and (iii) we show across Mandarin and English that refinement improves cross-model agreement and cross-speaker consistency, increases speaker similarity in TTS and VC, and maintains competitive intelligibility.

2. RELATED WORK

Invariant and efficient semantic speech tokenization. Semantic units are commonly obtained by quantizing self-supervised representations such as HuBERT [1]. To suppress speaker and noise variation, ContentVec disentangles speakers from continuous representations, R-Spin matches perturbed views, and PINT exploits parallel utterances and augmentations [3, 5, 6]. Duration variation is a separate source of inconsistency. TASTE uses text-aligned segment aggregation, while FlexiCodec learns adaptive frame rates to shorten token sequences [7, 8]. NormToken combines both perspectives through iterative semantic token purification: deduplication removes duration-induced repetitions, while T2U supervision suppresses residual variation that cannot be predicted from text. It requires neither speaker labels nor parallel same-content recordings. Thus, text predictability itself serves as the normalization criterion instead of explicitly supervising individual acoustic factors.

Text-to-unit Modeling. T2U models have been used to synthesize training data from text. BT4ST generates source speech through target-to-unit and unit-to-speech models, whereas DUB directly uses predicted source units to augment a unit-to-text translation model [9, 10]. Zeng et al. convert sampled text spans into speech-token spans to construct large-scale synthetic interleaved data for speech-text pre-training [11]. These methods keep the unit extractor fixed and use T2U predictions to train a downstream model. NormToken instead uses them to supervise the next iteration S2U tokenizer.

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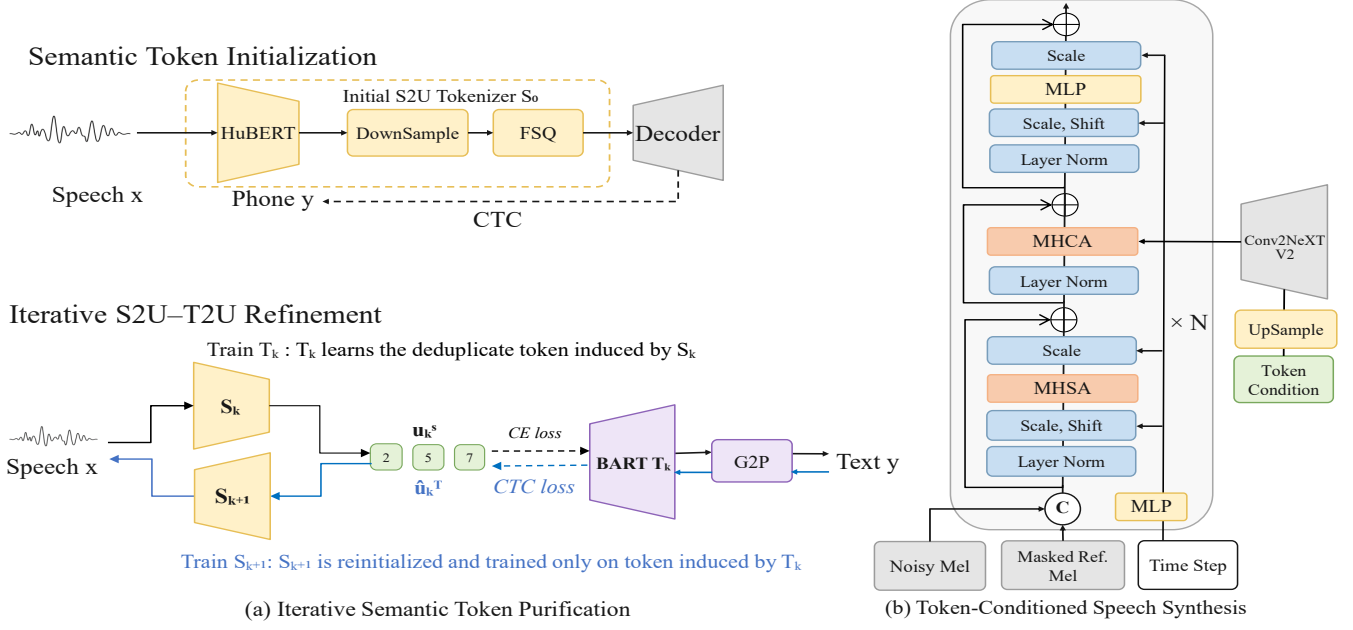


Fig. 1. Overview of the proposed framework. (a) NormToken performs iterative semantic token purification by alternating T2U training on deduplicated S2U sequences and S2U reinitialization using text-conditioned pseudo-targets. (b) The token-conditioned de-tokenizer synthesizes speech from either S2U or T2U tokens.

3. PROPOSED METHOD

3.1. Overview and Initialization

Figure 1 summarizes the full framework. Given paired speech and transcript $(\mathbf{x}, \mathbf{y})$, NormToken alternates an S2U tokenizer S_k and a T2U model T_k . Here, k denotes the refinement iteration rather than joint optimization: S_k is established before T_k is trained, and T_k then provides the supervision for S_{k+1} .

The initial tokenizer S_0 follows the semantic branch of DSA-Tokenizer [12]. A HuBERT encoder [1] extracts contextual speech representations, which are temporally downsampled and quantized using finite scalar quantization (FSQ) [13]. FSQ uses six scalar dimensions with four levels per dimension, yielding $4^6 = 4,096$ discrete units. The tokenizer operates at 25 Hz. During initial training, an auxiliary phoneme CTC decoder encourages the quantized representations to preserve linguistic content. This decoder is discarded after S_0 is trained; only the discrete FSQ indices are used to initialize the alternating refinement procedure.

3.2. Iterative S2U–T2U Refinement

At iteration k , consecutive repetitions in the current S2U sequence are removed by a deduplication operator $\mathcal{D}$. Deduplication converts the 25-Hz frame-level output into a shorter target sequence while preserving the order of unit transitions. A BART encoder-decoder [14] takes the phoneme sequence obtained by grapheme-to-phoneme (G2P) conversion as input and is trained to predict this sequence:

$$\mathbf{u}_k^S = \mathcal{D}(S_k(\mathbf{x})), \quad \mathcal{L}_T^k = -\log p_{T_k}(\mathbf{u}_k^S | \text{G2P}(\mathbf{y})). \quad (1)$$

Thus, T_k first learns the token inventory and ordering induced by the current S_k . At inference, it autoregressively decodes a pseudo-target

$\hat{\mathbf{u}}_k^T$ from the transcript. Because this prediction is conditioned on text rather than the source waveform, it cannot directly access the source speaker or utterance-level timing.

We then newly initialize S_{k+1} rather than fine-tuning S_k . The new S2U model applies a CTC classifier directly to the downsampled HuBERT representations and uses the same 4,096-unit vocabulary. It is trained only on the pseudo-target generated by T_k :

$$\begin{aligned} \hat{\mathbf{u}}_k^T &= \text{Decode}(T_k(\text{G2P}(\mathbf{y}))), \\ \mathcal{L}_S^{k+1} &= -\log p_{\text{CTC}}(\hat{\mathbf{u}}_k^T | \mathbf{x}). \end{aligned} \quad (2)$$

CTC provides monotonic alignment without frame-level correspondence. The fresh initialization is important because S_{k+1} must relearn the speech-to-unit mapping from the text-constrained targets instead of directly inheriting the previous classifier. Once training finishes, the output of S_{k+1} is decoded and deduplicated to train T_{k+1} , and the same two-step procedure repeats. Since every transfer from T2U to S2U passes through a model conditioned only on text, the cycle favors unit patterns that are consistently recoverable from linguistic content, while progressively reducing reliance on utterance-specific speaker and duration cues.

The two refinement steps play complementary roles. Deduplication removes duration variation explicitly represented by consecutive repetitions, whereas T2U supervision further suppresses token variations that cannot be predicted from the transcript. The alternating procedure can therefore be viewed as moving S2U and T2U toward a shared, text-predictable token space. Their convergence across iterations provides an empirical indication that this token space is approaching a stable representation of linguistic content.

3.3. Token-Conditioned Speech Synthesis

To test generation utility, we train separate de-tokenizers with the same architecture for iteration 0 and the final iteration K . This

avoids penalizing refined tokens merely because a decoder was fitted to the initial token distribution. Each de-tokenizer is trained on a mixture of S2U sequences extracted from speech and T2U sequences predicted from text at the corresponding iteration.

We adopt the CA-F5-TTS de-tokenizer used in Speech-Omni-Lite [15, 16]. The discrete unit sequence is upsampled and supplied as the semantic condition, while a masked reference mel spectrogram provides acoustic context. A conditional flow matching objective reconstructs the target mel spectrogram. At evaluation, S2U tokens extracted from a source utterance are used for VC, whereas T2U tokens predicted from text are used for TTS. The shared de-tokenizer design therefore tests both speech-derived and text-predicted units under the same synthesis backbone.

4. EXPERIMENTS

4.1. Experimental Setup

Training data. We sample approximately 8,000 hours of paired speech and text. The English portion draws from LibriSpeech [17], GigaSpeech [18], Libri-Heavy [19], Common Voice [20], and The People’s Speech [21]. The Mandarin portion draws from AISHELL-2 [22], WenetSpeech [23], and MagicData-RAMC [24]. The same multilingual pool is used for both S2U and T2U training. For T2U, each transcript is converted by G2P into a phoneme sequence, which serves as input; the deduplicated unit sequence produced by the corresponding S2U tokenizer is the target. Each refined S2U tokenizer is newly initialized and trained from the T2U pseudo-targets of the preceding iteration. The de-tokenizers for NormToken (Iter. 0) and NormToken (Iter. 4) are trained separately with the same architecture and the same approximately 100,000 hours of Mandarin and English speech from Emilia [25]. Each de-tokenizer uses a mixture of S2U- and T2U-derived sequences from its corresponding token space.

Baselines. For generation, we compare with MaskGCT [26], CosyVoice 2 [27], and CosyVoice 3 [28]. We also include DualCodec with MaskGCT [29], TaDiCodec-AR [30], Spark-TTS [31], and X-VC [32]. For token consistency, we include StableToken [33], SAC [34], FlexiCodec [8], R-Spin [5], and the semantic tokenizer of CosyVoice 3. Table 1 takes all TTS baseline results and the X-VC result from their original papers; the other VC results are evaluated by us.

Evaluation metrics. We use token-sequence WER and BLEU for cross-model agreement, and English WER, Chinese CER, and speaker similarity (SIM) on Seed-TTS [35]. On VCTK [36], UED [37] and SelfBLEU-4 measure consistency among utterances sharing the same sentence. For UED, we average pairwise edit distances among recordings of the same sentence, normalizing each distance by the reference sequence length. SelfBLEU-4 treats each sequence as the hypothesis and the other same-sentence sequences as references, computes sentence BLEU with uniform 1–4-gram weights and method-1 smoothing, and then averages over utterances and sentences. Speaking rate is computed for each utterance as its phone count divided by its duration in seconds.

4.2. Iterative Cross-Model Alignment

At each T_k point, WER and BLEU compare the predictions of T_k with the sequences produced by S_k , while each S_{k+1} point compares the outputs of S_{k+1} with the pseudo-targets generated by T_k . Figure 2 shows that most of the alignment gain occurs in the first two refinement rounds, after which both metrics approach a plateau.

Table 1. Multi-reference VC and TTS results on Seed-TTS. Best results are bolded; second-best results are underlined.

Model	English		Chinese	
	WER ↓	SIM ↑	CER ↓	SIM ↑
<i>Text-to-Speech</i>				
MaskGCT	2.62	<u>0.71</u>	2.27	<u>0.77</u>
CosyVoice2	2.57	0.65	1.45	0.75
CosyVoice3	2.02	0.72	<u>1.16</u>	0.78
DualCodec-MaskGCT	4.18	0.68	<u>1.85</u>	0.75
TaDiCodec-AR-0.5B	3.88	0.65	1.15	0.75
Spark-TTS	1.98	0.58	1.20	0.67
NormToken(Iter. 0)	1.80	0.63	1.52	0.72
NormToken(Iter. 4)	<u>1.97</u>	<u>0.71</u>	1.78	0.78
<i>Voice Cloning</i>				
CosyVoice2	3.48	0.52	2.91	0.72
CosyVoice3	2.74	0.56	2.21	<u>0.75</u>
X-VC	3.14	0.62	<u>2.65</u>	0.72
NormToken(Iter. 0)	3.00	0.63	<u>2.65</u>	0.73
NormToken(Iter. 4)	<u>2.94</u>	0.71	2.76	0.77

The largest jump follows the first T2U update, suggesting that text-conditioned prediction filters token variations not consistently recoverable from linguistic content. The subsequent improvement of the newly initialized S_2 trained on T_1 pseudo-targets shows that this gain transfers back to S2U. More generally, the improvement obtained by each T_k is largely retained when its pseudo-targets supervise the freshly initialized S_{k+1} . This recurring transfer indicates that refinement changes the S2U mapping rather than only improving the T2U predictor. The similar trajectories on all four test sets also show that this behavior is consistent across languages and corpus conditions. From T_0 to T_4 , WER decreases by 72.0–86.7% relative, while BLEU increases by 58.88–73.39 points. By iteration 4, S2U and T2U differ by at most 0.0058 WER and 0.85 BLEU, indicating a stable shared token space across languages and domains. Since agreement alone does not establish token quality, the following experiments assess generation utility and utterance-specific information.

4.3. Generation Utility

The two tasks evaluate complementary paths through the token space. VC uses speech-derived S2U tokens, whereas TTS uses text-predicted T2U tokens. Their joint performance therefore tests whether both sides of the iterative cycle remain useful for generation. Table 1 shows that both token spaces support competitive TTS and VC. For TTS, NormToken (Iter. 0) gives the best English WER (1.80) among all baseline systems. NormToken (Iter. 4) ranks second with a slightly higher WER of 1.97, while improving speaker similarity from 0.63 to 0.71. It also reaches the joint-best Chinese speaker similarity of 0.78. For VC, NormToken (Iter. 4) obtains the best speaker similarity in both languages and the second-best English WER, raising SIM over Iter. 0 from 0.63 to 0.71 in English and from 0.73 to 0.77 in Chinese. Since both de-tokenizers use the same architecture and Emilia training data, the stronger SIM at Iter. 4 indicates that refined tokens better support reference-conditioned speaker realization. Meanwhile, its competitive WER and CER confirm that iterative refinement preserves content representation. Overall, refinement strengthens speaker-normalized generation without sacrificing the linguistic information required by VC and TTS. The simultaneous SIM gains suggest a cleaner division

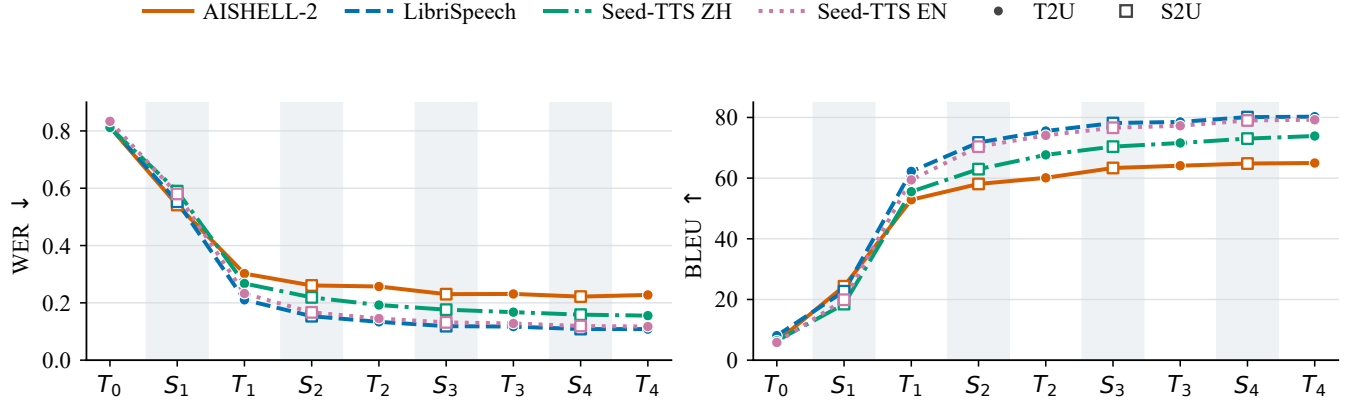


Fig. 2. WER and BLEU agreement over iterative S2U-T2U refinement on four test sets.

Table 2. Token consistency across parallel VCTK utterances. SelfBLEU-4 is reported on a 0–100 scale.

Tokenizer	UED ↓	SelfBLEU-4 ↑
StableToken	344.54	30.95
SAC semantic	409.72	4.37
CosyVoice 3	404.70	10.52
FlexiCodec (12.5 Hz)	428.93	2.37
R-Spin (2048)	241.42	76.64
Contentvec	320.96	40.11
NormToken(Iter. 0)	344.61	27.04
NormToken(Iter. 4)	59.44	94.17

of labor: semantic tokens carry content, while the acoustic reference supplies speaker identity.

4.4. Parallel-Utterance Token Consistency

We group VCTK utterances by shared sentence. Each group contains recordings from the subset of speakers who read that sentence, so the number of speakers varies across groups; it does not include the full VCTK speaker set. Relative to S_0 , S_4 reduces UED by 82.7% and raises SelfBLEU-4 by 67.13 points. Its SelfBLEU-4 score is substantially higher than those of the listed baselines, showing that iterative refinement makes parallel utterances converge to much more similar token sequences. UED captures global sequence edits, whereas SelfBLEU-4 measures local token overlap. Improving both indicates that refinement aligns global ordering and recurring transitions. These consistency gains are not explained by a trivial collapse to a small set of uninformative tokens. Such a collapse would remove distinctions needed to recover the utterance content, whereas NormToken (Iter. 4) retains competitive WER and CER in Table 1. The joint improvements in consistency and generation therefore indicate that refinement removes utterance-specific variation while preserving linguistically useful contrasts.

4.5. Reference-Conditioned Speaking Rate

In VC, the source utterance provides linguistic content, while the acoustic reference provides the target-speaker condition. Figure 3 shows that the speaking rates of CosyVoice 3 and NormToken

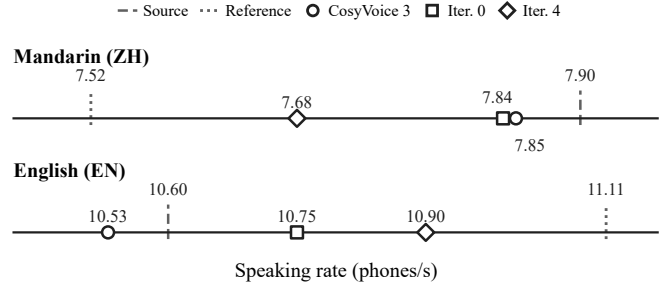


Fig. 3. VC speaking rates relative to the semantic source and acoustic reference.

(Iter. 0) remain closer to the source in both languages. After refinement, this pattern reverses: the speaking rate produced by NormToken (Iter. 4) is closer to the acoustic reference than to the source utterance. This reversal suggests that the refined tokens retain less source-side duration information, making the generated speech less dependent on the source speaking rate and supporting the intended duration-normalization effect.

5. CONCLUSION

NormToken addresses inconsistency in discrete semantic tokens through iterative S2U-T2U alignment, progressively driving both models toward a shared, text-predictable token space. Across Mandarin and English, refinement improves cross-model agreement and parallel-utterance consistency while reducing cross-speaker sequence variation and duration dependence. Importantly, NormToken (Iter. 4) improves SIM over NormToken (Iter. 0) for both TTS and VC in both languages while retaining competitive WER and CER. Thus, NormToken preserves linguistic content while better supporting reference-conditioned speaker similarity and speaking-rate transfer, yielding more effective semantic tokens for speech generation.

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